Week 11: November 9-13
Chemistry in Biology

Subject/:الموضوع Biology **Unit Title: عنوان الوحدة The Cell** **Gradeالصف : 9**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1	I can explain how the plasma membrane maintains cellular homeostasis by modelling the phospholipid bilayer, identifying the functions of membrane components, and relating selective permeability to the movement of substances into and out of the cell.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- The Plasma Membrane.pdf
Lesson 2	I can explain and compare passive transport mechanisms by interpreting concentration gradients, modelling diffusion and facilitated diffusion, and predicting the movement of water through a selectively permeable membrane.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Cellular Transport.pdf
Lesson 3	I can predict and explain the effects of osmosis on plant and animal cells by comparing isotonic, hypotonic, and hypertonic solutions, interpreting concentration differences, and drawing arrows to represent the net movement of water.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Cellular Transport.pdf
Lesson 4	I can compare passive and active transport and explain how cells move large particles by modelling movement against concentration gradients, identifying energy requirements, and illustrating endocytosis and exocytosis.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Cellular Transport.pdf
Homework الواجبات			

Teacher's notes/ Reminders ملاحظات المعلم والتذكير	

Week # 11 الأسبوع#

Technology Integration دمج التكنولوجيا	Digital fluid mosaic model Digital particle simulation Interactive osmosis simulation
Cross-curricular links الروابط بين المواد الدراسية	Chemistry: Polar and nonpolar molecules, phospholipid structure, and interactions with water. Physics: Movement and organization of particles within a flexible structure. Chemistry: Concentration, dissolved particles, and movement of substances in solutions. Physics: Random motion, temperature, particle collisions, and pressure. Chemistry: Concentration of dissolved substances in solutions. Mathematics: Comparing relative concentrations and interpreting changes in cell size.
External Assessment Integration دمج التقييمات الخارجية	NA
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Research & Thinking (ATL): Relate membrane structure to function and predict the consequences of changes to membrane components. Communication: Use scientific vocabulary to describe the phospholipid bilayer and explain the roles of membrane components. Collaboration: Construct a shared membrane model and agree on the correct placement and function of each component. Research & Thinking (ATL): Interpret particle distributions, identify concentration gradients, and use evidence to predict the direction of movement. Communication: Explain transport mechanisms using precise scientific vocabulary. Collaboration: Compare particle models and discuss differences between diffusion, facilitated diffusion, and osmosis.

Standards المعايير	HS-LS1-6: Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules may combine with other elements to form amino acids and/or other large carbon-based molecules.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How do scientific models help us understand structures that are too small to observe directly? How can models of particle movement help us understand processes that cannot be directly observed inside living cells?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 MUST DOK 1 – Recall: Identify the components of the plasma membrane and state their functions. SHOULD DOK 2/3 – Apply & Analyze: Explain how phospholipid arrangement and membrane components contribute to selective permeability and homeostasis. COULD DOK 4 – Evaluate & Create: Construct and justify a labelled plasma membrane model and predict the consequences of removing or damaging one component. Lesson 2 MUST DOK 1 – Recall: Define diffusion, facilitated diffusion, osmosis, passive transport, and dynamic equilibrium. SHOULD DOK 2/3 – Apply & Analyze: Interpret concentration gradients, identify the direction of net movement, and compare diffusion, facilitated diffusion, and osmosis. COULD DOK 4 – Evaluate & Create: Construct and justify a model comparing the three passive transport mechanisms, including movement direction, membrane requirements, and energy requirements. Lesson 3 MUST DOK 1 – Recall: Define isotonic, hypotonic, and hypertonic solutions. SHOULD DOK 2/3 – Apply & Analyze: Identify the type of solution, draw the net movement of water, and explain the effects on plant and animal cells. COULD DOK 4 – Evaluate & Create: Construct and justify a model showing how changes in surrounding water availability affect plant and animal cells, and propose a suitable response to a plant water-stress scenario. Lesson 4 MUST DOK 1 – Recall: Define active transport, endocytosis, and exocytosis.

	SHOULD DOK 2/3 – Apply & Analyze: Compare passive and active transport and explain the steps involved in endocytosis and exocytosis. COULD DOK 4 – Evaluate & Create: Develop and justify a transport decision flowchart that classifies membrane transport mechanisms according to movement direction, particle size, membrane structures, and energy requirements.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Citizenship — Conservation Values – Global understanding
	Area: Responsible water use, environmental quality, and protection of living organisms in UAE ecosystems. Area: Water balance, survival of organisms, and conservation of UAE desert and coastal ecosystems.
	How to Implement: Connect selective permeability and homeostasis to the ability of organisms to survive in changing environments. Discuss how organisms living in UAE coastal and desert environments must regulate the movement of water and other substances to maintain internal balance. Students explain why cellular regulation is essential for survival. How to Implement: Discuss how cells must regulate water movement to survive in environments with different water availability. Connect osmosis to the challenges faced by plants and other organisms in UAE desert and coastal environments. Students explain why water balance is essential for cellular survival. How to Implement: Connect osmosis and plant cell water balance to the challenges of growing plants in the UAE. Students investigate why water conservation and suitable irrigation practices are important for maintaining plant health. They explain how a shortage of available water can contribute to wilting and propose responsible water-use practices.